

Syllabus - BTCS0201 Data Structures

Sr. No	Module/Units	Detailed Topic wise Syllabus	Total Hours (L+T+P+S)
1	Unit 1: Introduction to Algorithms and Data structures	Concepts and design of algorithm Various notations used in representing complexities of algorithms Introduction to data structures and their utilities Types of data structures Data structure operations	
2	Unit 2: Arrays, Sorting and Searching	Introduction and types of arrays. Representation of one dimensional arrays in memory. Array traversal. Insertion and deletion in an array. Basic sorting and searching using arrays. Representation of multi-dimensional arrays in memory. Realizing matrices using two dimensional arrays. Various popular sorting techniques with their complexity analysis. Various popular searching. Techniques with their complexity analysis.	
3	Unit 3: Stacks and queues	Introduction of stacks, Stack operations, Stack implementation using arrays, Practical utility of stacks, Introduction of queues, Queue operations, Queue implementation using arrays. Different kinds of queues	
4	Unit 4: Linked List	Introduction and utility of linked lists Concepts and implementation of linked list using dynamic memory allocation, Types of linked lists, Circular linked list, Double linked list, Practical applications of linked list	
5	Unit 5: Trees, Graphs and Practical applications of data structures	Introduction and basic concept of trees Binary Tree, Binary tree traversal, Binary search tree, Tree variants Basic concepts and terminologies used in graphs. Graph implementation, Graph traversal, Shortest path algorithm Mini project involving various data structure.	